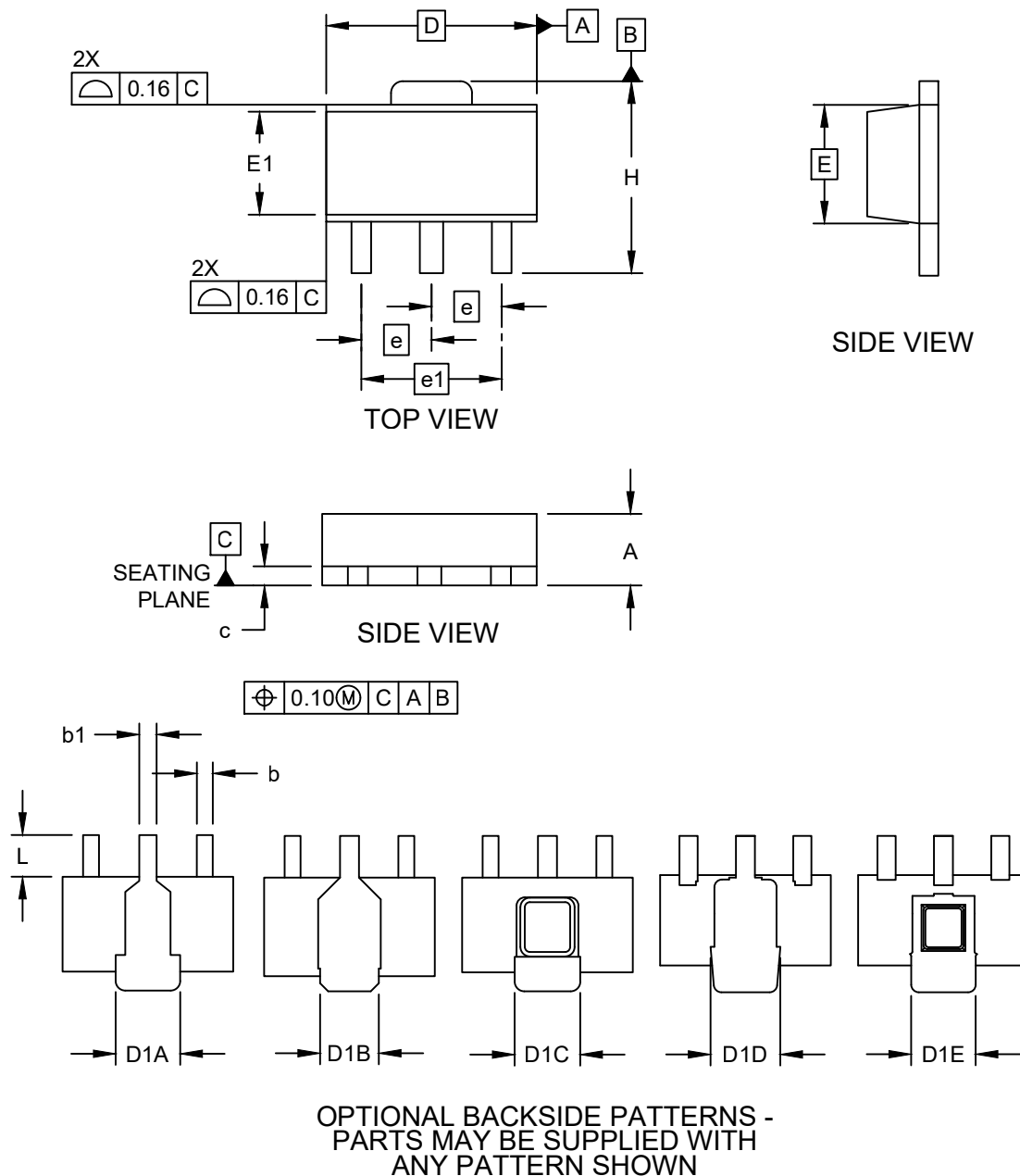


Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (MB) - [SOT-89]

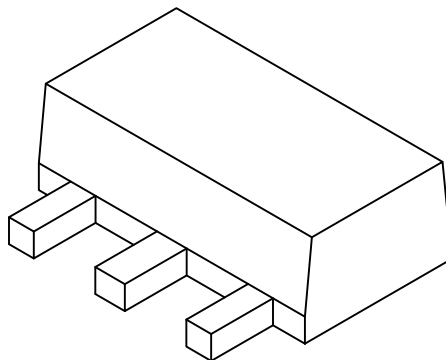
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (MB) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Leads	N	3		
Pitch	e	1.50 BSC		
Outside Lead Pitch	e1	3.00 BSC		
Overall Height	A	1.40	1.50	1.60
Overall Width	H	3.94	4.10	4.25
Molded Package Width at Base	E	2.50 BSC		
Molded Package Width at Top	E1	2.00	2.20	2.29
Overall Length	D	4.50 BSC		
Tab Length (Option A)	D1A	1.63	1.73	1.83
Tab Length (Option B)	D1B	1.40	1.60	1.75
Tab Length (Option C)	D1C	1.62	1.73	1.83
Tab Length (Option D)	D1D	1.64 BSC		
Tab Length (Option E)	D1E	1.62	1.73	1.83
Foot Length	L	0.73	1.10	1.20
Lead Thickness	c	0.35	0.40	0.44
Leads 1 & 3 Width	b	0.36	0.42	0.48
Lead 2 Width	b1	0.41	0.47	0.56
Leads 1, 2, & 3 Width (Option D)	b	0.43	0.48	0.54

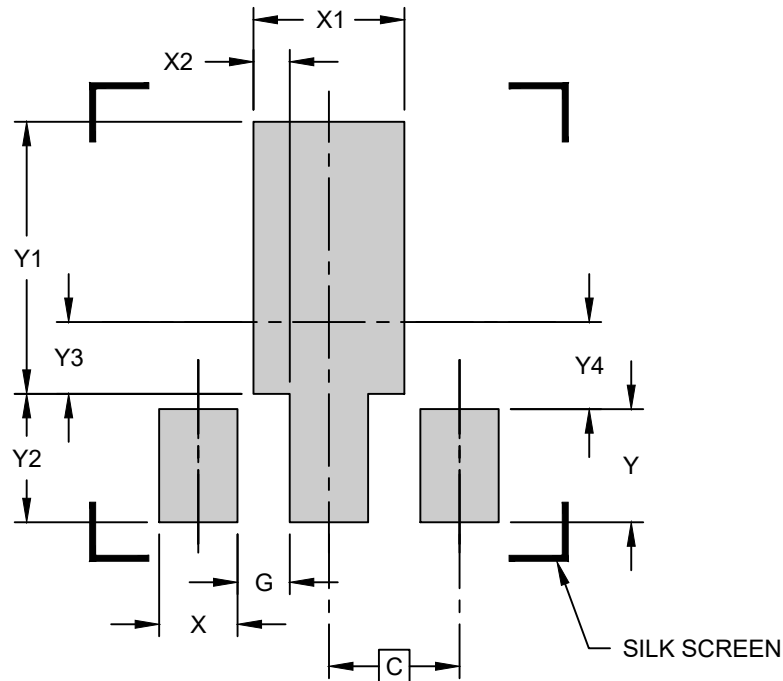
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

3-Lead Plastic Small Outline Transistor (MB) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units	MILLIMETERS		
Dimension Limits	MIN	NOM	MAX
C	1.50 (BSC)		
X (3 PLACES)		0.900	
X1		1.733	
X2 (2 PLACES)		0.416	
G (2 PLACES)		0.600	
Y (2 PLACES)		1.300	
Y1		3.125	
Y2		1.475	
Y3		0.825	
Y4		1.000	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

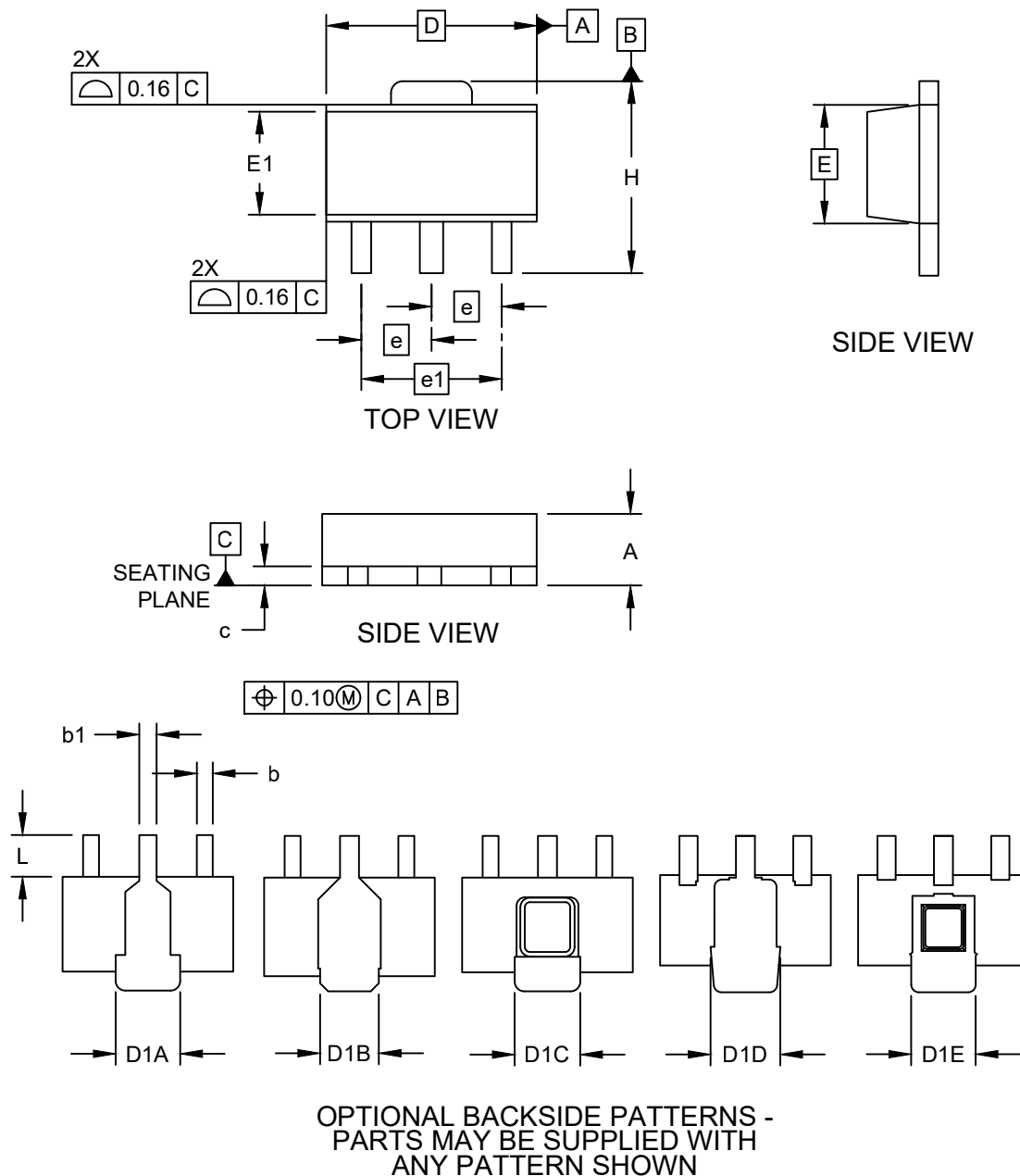
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-02029-MB Rev. E

Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (2AX) - [SOT-89]

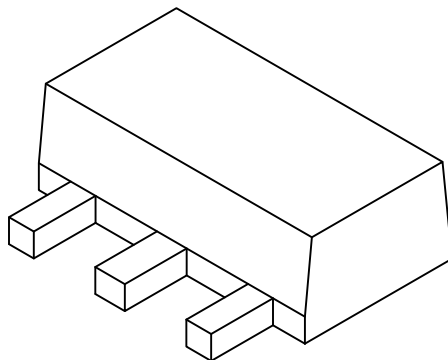
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (2AX) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Leads	N	3		
Pitch	e	1.50 BSC		
Outside Lead Pitch	e1	3.00 BSC		
Overall Height	A	1.40	1.50	1.60
Overall Width	H	3.94	4.10	4.25
Molded Package Width at Base	E	2.50 BSC		
Molded Package Width at Top	E1	2.00	2.20	2.29
Overall Length	D	4.50 BSC		
Tab Length (Option A)	D1A	1.63	1.73	1.83
Tab Length (Option B)	D1B	1.40	1.60	1.75
Tab Length (Option C)	D1C	1.62	1.73	1.83
Tab Length (Option D)	D1D	1.64 BSC		
Tab Length (Option E)	D1E	1.62	1.73	1.83
Foot Length	L	0.73	1.10	1.20
Lead Thickness	c	0.35	0.40	0.44
Leads 1 & 3 Width	b	0.36	0.42	0.48
Lead 2 Width	b1	0.41	0.47	0.56
Leads 1, 2, & 3 Width (Option D)	b	0.43	0.48	0.54

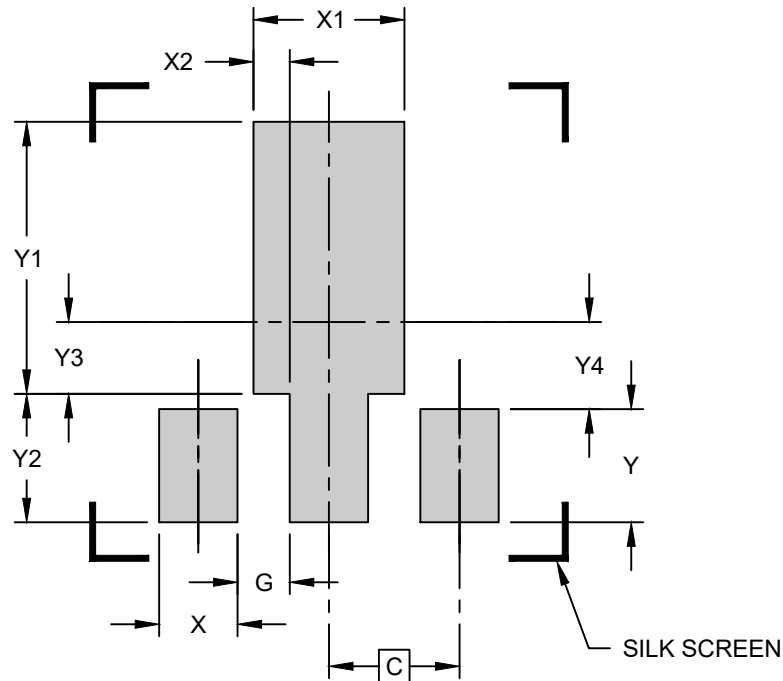
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

3-Lead Plastic Small Outline Transistor (2AX) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units	MILLIMETERS		
Dimension Limits	MIN	NOM	MAX
C	1.50 (BSC)		
X (3 PLACES)		0.900	
X1		1.733	
X2 (2 PLACES)		0.416	
G (2 PLACES)		0.600	
Y (2 PLACES)		1.300	
Y1		3.125	
Y2		1.475	
Y3		0.825	
Y4		1.000	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

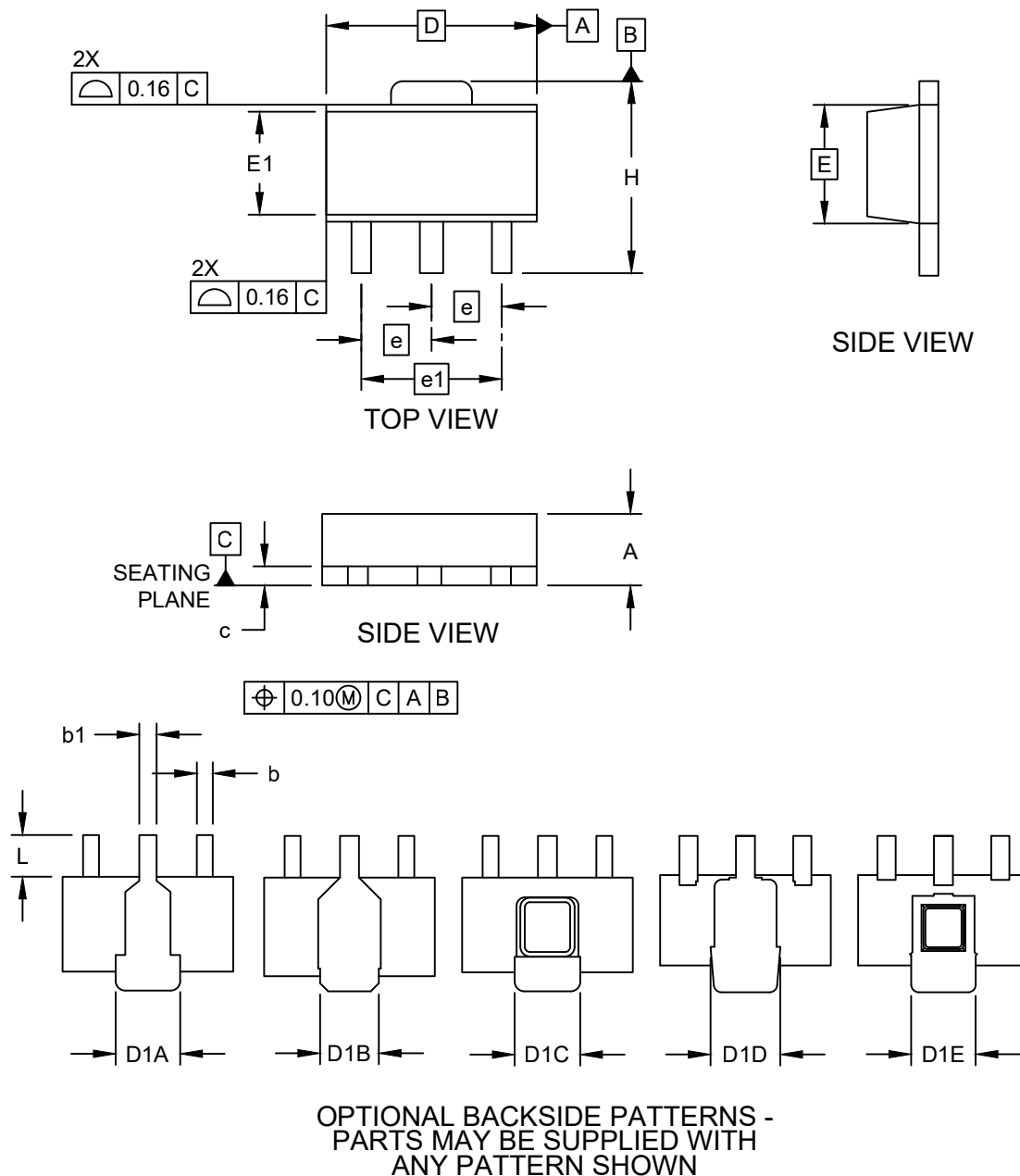
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-02029-2AX Rev. E

Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (A5X) - [SOT-89]

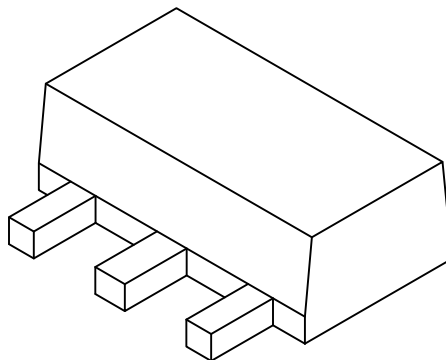
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (A5X) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Leads	N	3		
Pitch	e	1.50 BSC		
Outside Lead Pitch	e1	3.00 BSC		
Overall Height	A	1.40	1.50	1.60
Overall Width	H	3.94	4.10	4.25
Molded Package Width at Base	E	2.50 BSC		
Molded Package Width at Top	E1	2.00	2.20	2.29
Overall Length	D	4.50 BSC		
Tab Length (Option A)	D1A	1.63	1.73	1.83
Tab Length (Option B)	D1B	1.40	1.60	1.75
Tab Length (Option C)	D1C	1.62	1.73	1.83
Tab Length (Option D)	D1D	1.64 BSC		
Tab Length (Option E)	D1E	1.62	1.73	1.83
Foot Length	L	0.73	1.10	1.20
Lead Thickness	c	0.35	0.40	0.44
Leads 1 & 3 Width	b	0.36	0.42	0.48
Lead 2 Width	b1	0.41	0.47	0.56
Leads 1, 2, & 3 Width (Option D)	b	0.43	0.48	0.54

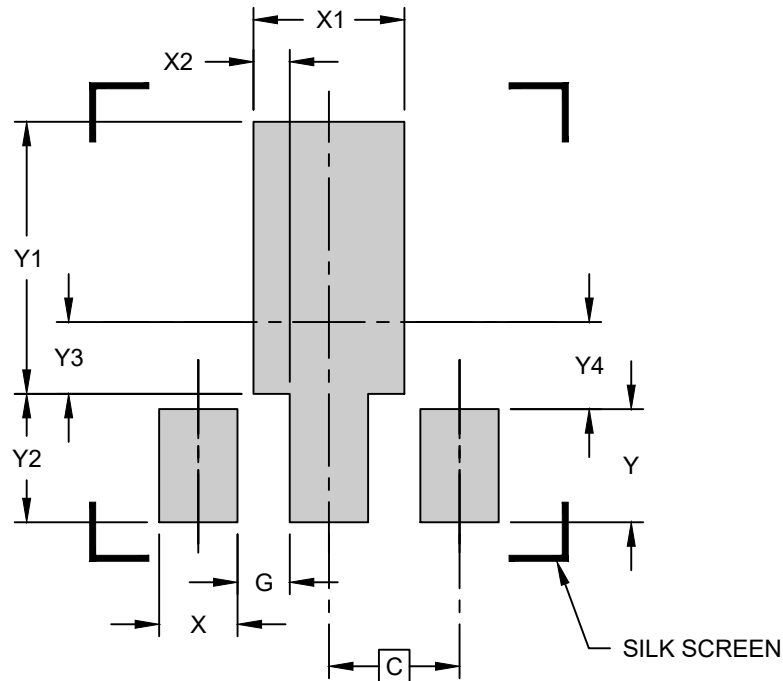
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

3-Lead Plastic Small Outline Transistor (A5X) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units	MILLIMETERS		
Dimension Limits	MIN	NOM	MAX
C	1.50 (BSC)		
X (3 PLACES)		0.900	
X1		1.733	
X2 (2 PLACES)		0.416	
G (2 PLACES)		0.600	
Y (2 PLACES)		1.300	
Y1		3.125	
Y2		1.475	
Y3		0.825	
Y4		1.000	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

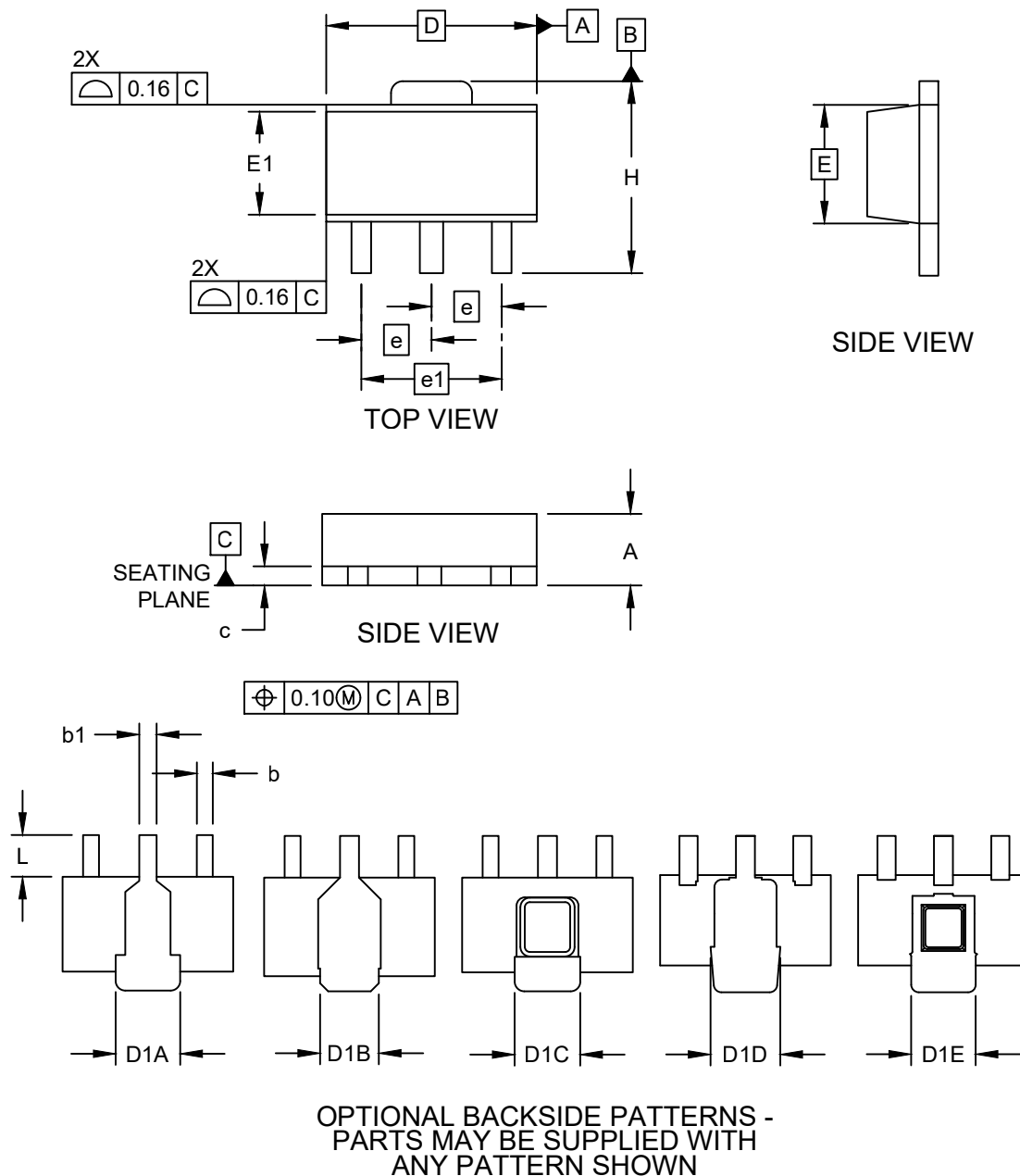
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-02029-A5X Rev. E

Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (ATX) - [SOT-89]

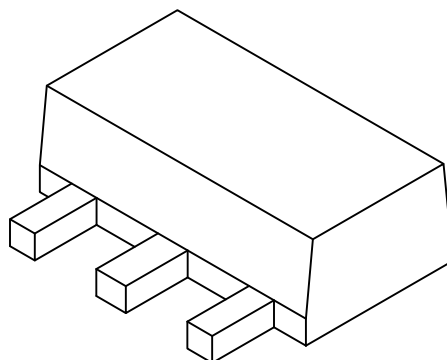
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (ATX) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Leads	N	3		
Pitch	e	1.50 BSC		
Outside Lead Pitch	e1	3.00 BSC		
Overall Height	A	1.40	1.50	1.60
Overall Width	H	3.94	4.10	4.25
Molded Package Width at Base	E	2.50 BSC		
Molded Package Width at Top	E1	2.00	2.20	2.29
Overall Length	D	4.50 BSC		
Tab Length (Option A)	D1A	1.63	1.73	1.83
Tab Length (Option B)	D1B	1.40	1.60	1.75
Tab Length (Option C)	D1C	1.62	1.73	1.83
Tab Length (Option D)	D1D	1.64 BSC		
Tab Length (Option E)	D1E	1.62	1.73	1.83
Foot Length	L	0.73	1.10	1.20
Lead Thickness	c	0.35	0.40	0.44
Leads 1 & 3 Width	b	0.36	0.42	0.48
Lead 2 Width	b1	0.41	0.47	0.56
Leads 1, 2, & 3 Width (Option D)	b	0.43	0.48	0.54

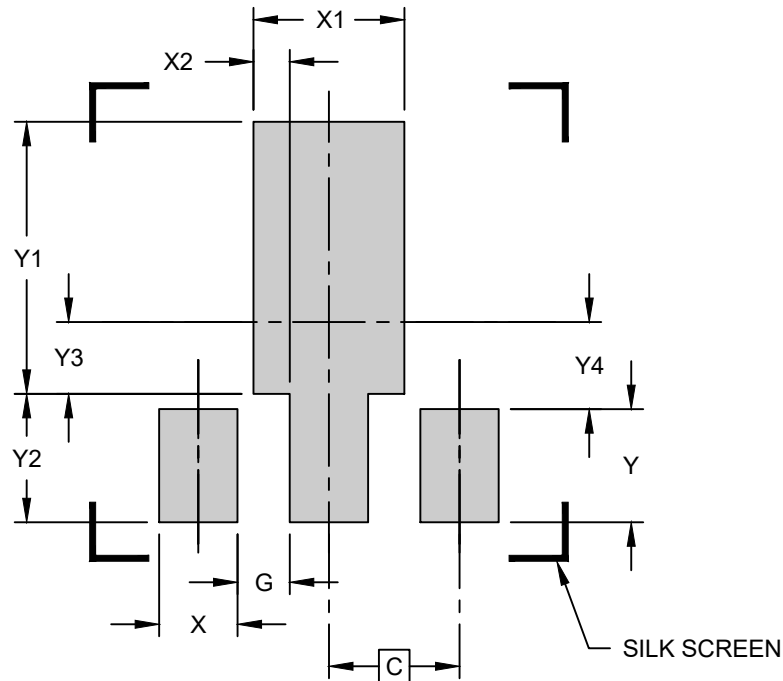
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

3-Lead Plastic Small Outline Transistor (ATX) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units	MILLIMETERS		
Dimension Limits	MIN	NOM	MAX
C	1.50 (BSC)		
X (3 PLACES)		0.900	
X1		1.733	
X2 (2 PLACES)		0.416	
G (2 PLACES)		0.600	
Y (2 PLACES)		1.300	
Y1		3.125	
Y2		1.475	
Y3		0.825	
Y4		1.000	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

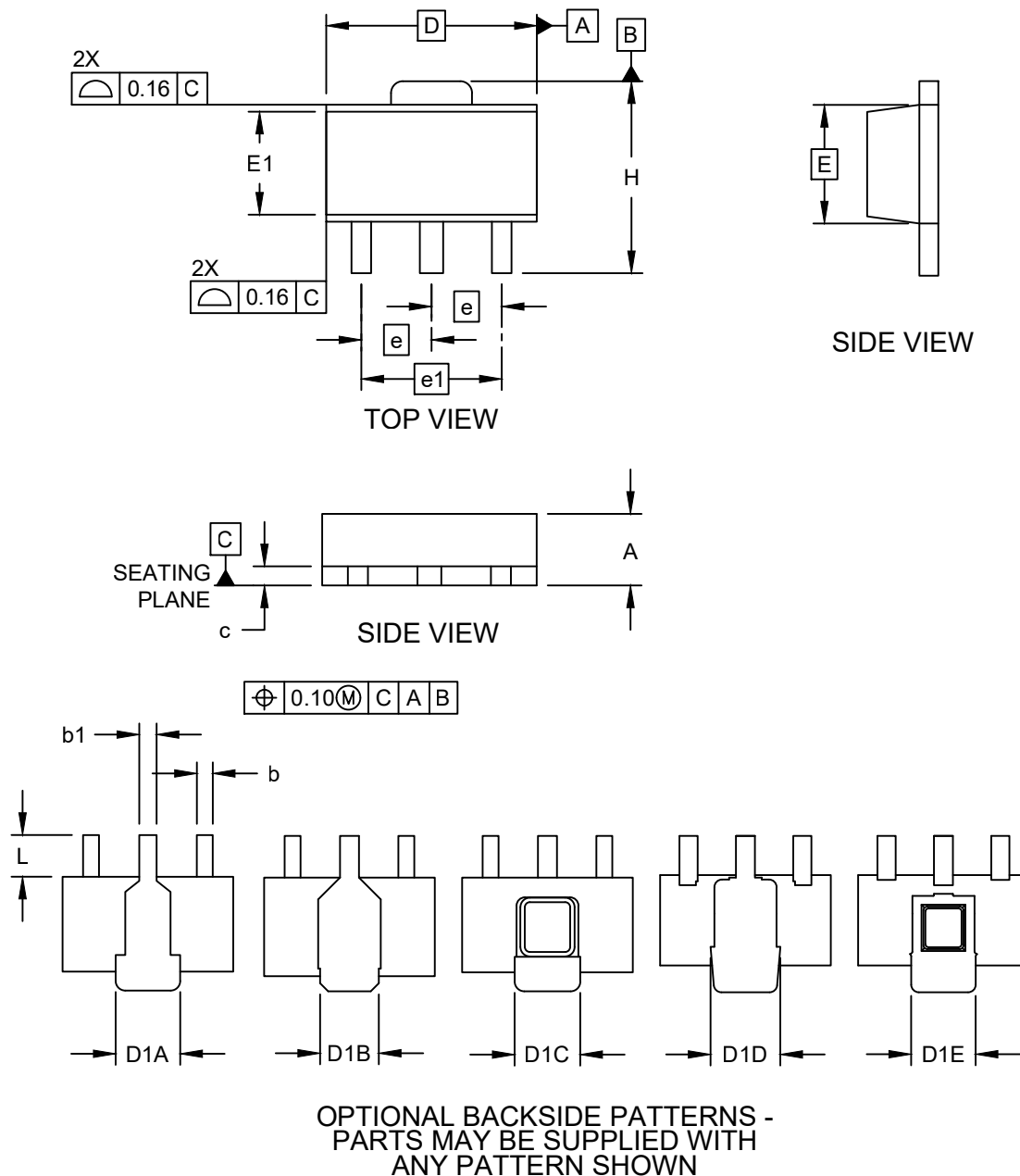
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-02029-ATX Rev. E

Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (BTX) - [SOT-89]

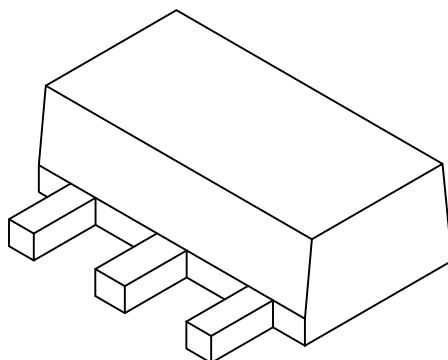
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

3-Lead Plastic Small Outline Transistor (BTX) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Leads	N	3		
Pitch	e	1.50 BSC		
Outside Lead Pitch	e1	3.00 BSC		
Overall Height	A	1.40	1.50	1.60
Overall Width	H	3.94	4.10	4.25
Molded Package Width at Base	E	2.50 BSC		
Molded Package Width at Top	E1	2.00	2.20	2.29
Overall Length	D	4.50 BSC		
Tab Length (Option A)	D1A	1.63	1.73	1.83
Tab Length (Option B)	D1B	1.40	1.60	1.75
Tab Length (Option C)	D1C	1.62	1.73	1.83
Tab Length (Option D)	D1D	1.64 BSC		
Tab Length (Option E)	D1E	1.62	1.73	1.83
Foot Length	L	0.73	1.10	1.20
Lead Thickness	c	0.35	0.40	0.44
Leads 1 & 3 Width	b	0.36	0.42	0.48
Lead 2 Width	b1	0.41	0.47	0.56
Leads 1, 2, & 3 Width (Option D)	b	0.43	0.48	0.54

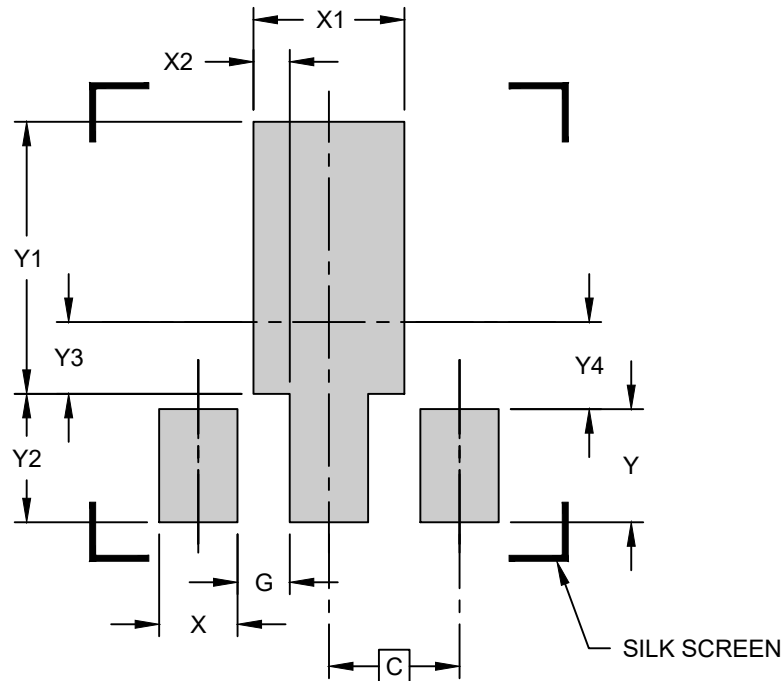
Notes:

1. The Pin 1 visual index feature may vary, but it must be located within the hatched area.
2. The package is saw singulated.
3. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

3-Lead Plastic Small Outline Transistor (BTX) - [SOT-89]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units	MILLIMETERS		
Dimension Limits	MIN	NOM	MAX
C	1.50 (BSC)		
X (3 PLACES)		0.900	
X1		1.733	
X2 (2 PLACES)		0.416	
G (2 PLACES)		0.600	
Y (2 PLACES)		1.300	
Y1		3.125	
Y2		1.475	
Y3		0.825	
Y4		1.000	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-02029-BTX Rev. E